Shikang (Danny) Wu

Senior Staff ML Scientist | Search & Recommendation

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Summary

Senior Staff ML Scientist focused on personalized ranking and generative search. Built and deployed large recommendation models and semantic-ID-based generative search systems at Douyin, including OneModel for unified retrieval and pre-ranking. Leads technical direction and partners with multimodal, search-engine, and ML infrastructure teams to take models from research to production and improve training and serving efficiency.

Industry Experience

ByteDance Senior Staff ML Scientist / Tech Lead Dec. 2023 – Present

Lead a 12-person team and set the technical direction for Douyin Search's core ranking models. Built and deployed large recommendation models and generative search models from the ground up to support personalized ranking and retrieval across multiple search scenarios and objectives. Received four "Exceed Expectations" ratings and one "Consistently Meet Expectations+" rating across five reviews.

Large Recommendation Model

Built and scaled personalized ranking models for Douyin Search, combining sparse architectures, unified token-based modeling, multimodal learning, and long-context user modeling.

- **Efficient Scaling:** Scaled dense parameters from **10M to 1B** and sparse parameters from **4T to 40T** by evolving DLRM into RankMixer and incorporating sparse MoE and gated memory retrieval. Improved embedding efficiency through parameter sharing, contrastive learning, and quantization to manage training and serving costs.
- **All-Token Modeling:** Developed MDL to jointly model heterogeneous features, search scenarios, and objectives through a unified token representation. Introduced HyFormer to enable direct interaction between sequence and non-sequence tokens, reducing information loss from two-stage sequence compression.
- **Multimodal Co-Training:** Moved from separately trained LLM/VLM representations to end-to-end co-training with personalized ranking models. Incorporated visual features and token-level query–document interactions, and enhanced user, query, and document representations through embedding write-back, aggregation, and caching.
- **Long-Context Modeling:** Enabled end-to-end modeling of **5K user interactions** and developed two-stage retrieval over **10K–100K historical interactions**, using SID-based filtering followed by multimodal embedding retrieval to select relevant user history.

Generative Search Model

Built a generative search system based on multimodal semantic IDs, bringing generation and pre-ranking into a unified model through two phases of production deployment.

- **Semantic Tokenization and Evaluation:** Established a search-specific evaluation framework for semantic IDs and partnered with the multimodal platform team to develop semantic tokens. Connected offline representation and candidate-quality evaluation with online search outcomes.
- **Model Development:** Scaled the task model and improved its instruction-following and multimodal capabilities to better capture user intent, query semantics, and content, enabling more controllable candidate generation.
- **Unified Retrieval and Pre-Ranking:** Deployed OneModel to jointly support retrieval and pre-ranking, fully replacing the legacy single-tower pre-ranker.

Baidu Senior Algorithm Engineer Jun. 2021 – Nov. 2023

Led user behavior modeling and feature development for search ranking, integrating heterogeneous result types to improve engagement while preserving search satisfaction. Rated "Exceeds Expectations" in the 2022 annual performance review (top 3%).

- **User Feedback and Dwell-Time Modeling:** Modeled explicit clicks and implicit autoplay feedback separately to account for differences across result formats. Built impression-level and post-click dwell-time models, using sample reweighting and multi-objective classification to improve on single-target regression.
- **Debiasing:** Developed a framework to account for presentation, position, and scenario biases. Combined propensity-based correction with outcome modeling in a doubly robust approach, then fine-tuned on randomized-exposure data to reduce distribution mismatch.
- **Personalized Ranking:** Evolved the model architecture from Wide & Deep to DeepFM and MMoE, incorporating richer query, document, and user features alongside search-intent signals.

Selected Research

- **MSN (Co-first Author; KDD 2026):** Scales personalized recommendation through Product-Key Memory retrieval and gated feature interaction, balancing model capacity with training and inference efficiency.
- **TokenMixer-Large (Co-first Author; KDD 2026):** Combines improved token mixing and residual connections, auxiliary supervision, and sparse per-token MoE to stabilize deep ranking models, scaling to 7B parameters online and 15B offline.
- **MDL (Co-first Author; KDD 2026):** Tokenizes features, scenarios, and tasks into a unified space, using feature self-attention, domain–feature attention, and domain-fused aggregation for joint multi-scenario, multi-task recommendation.
- **HyFormer (Co-corresponding Author; SIGIR 2026):** Unifies sequence modeling and feature interaction through alternating global-query attention over behavioral histories and efficient token mixing.
- **LEMUR (Co-author; CIKM 2026):** Jointly trains multimodal and recommendation modules for objective alignment and real-time updates, using a memory bank to efficiently reuse historical multimodal features.

Education

Beijing Jiaotong University | Computer Science

Master of Engineering Sep. 2018 – Jun. 2021
Bachelor of Engineering Sep. 2014 – Jun. 2018

Expertise

Model Architecture: Transformers; long-context sequence modeling; personalized recommendation and ranking; LLM/VLM representations.

Production ML: Large-scale model training and serving; training/inference optimization; offline evaluation and online experimentation.

Programming and Frameworks: Python, C++, SQL; TensorFlow, PyTorch.